SUPPORTING ADULT GUIDANCE

Percentages as multipliers

Percentages, Ratio, Growth and Bounds · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Multiplier Line, two linked number lines show the same point as a percentage (above) and a multiplier (below), with 100% fixed at 1. Bars show the original amount and the result. Three modes: percentage of an amount, increase or decrease, and one number as a percentage of another. Prices are whole pounds and percentages whole numbers, and amounts are held in pence, so every value is exact. In increase and decrease mode the readout shows the two-step method beside the single multiplier. Seven questions and a stretch follow (why +10% then –10% gives × 0.99). Movement is quick multiplier matching; Reflection has true-or-not statements; Creativity sets a price and a change to meet briefs.

Driving Question: *What single number does the change?*

The mathematics, in plain terms

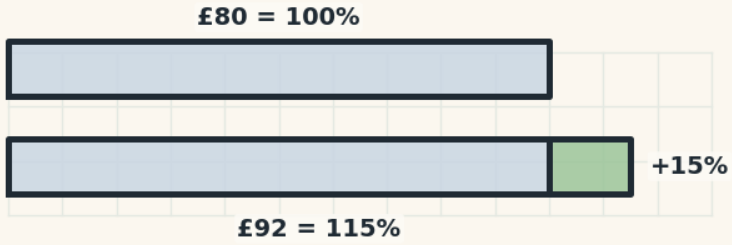
p% = p/100. Increasing by p% multiplies by (1 + p/100); decreasing by p% multiplies by (1 – p/100). A as a percentage of B is 100A/B. Successive percentage changes compose by multiplying their multipliers, so +10% then –10% is 1.1 × 0.9 = 0.99, a 1% decrease, because each percentage is of the amount current at that time.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Increase £80 by 15%

A jacket costs £80. Its price rises by 15%. What is the new price?

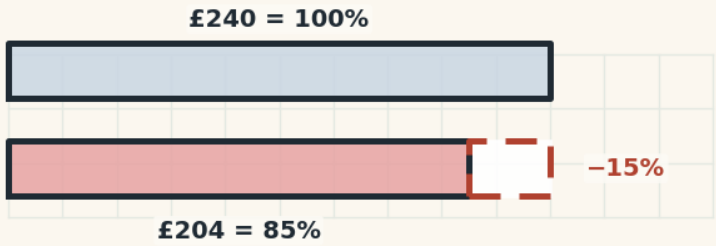


1. The new price is 100% + 15%	115% = 1.15
2. Multiply once	£80 × 1.15 = £92
3. Check with two steps	15% of £80 = £12; £80 + £12 = £92

The new price is £92.

Example 2 — Decrease £240 by 15%

A £240 phone is reduced by 15% in a sale. What is the sale price?



1. What is left: 100% − 15%	85% = 0.85
2. Multiply once	£240 × 0.85 = £204

The sale price is £204.

See the multiplier, then use it

Learners see a percentage and its multiplier at the same point on linked lines before using multipliers in questions. The two-step method stays visible beside the single multiplier, so the multiplier is a shortcut they understand, not a rule to remember.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Multiplier Line: percentage of, increase or decrease, as a percentage of. Seven questions. Stretch: up 10% then down 10%.	~16 min
Movement	Multiplier match.	~6 min
Reflection	True or not? × 0.2 versus a 20% decrease.	~8 min
Creativity	Price it: six briefs.	~10 min
Rest	One number does the whole job.	~2 min
Nutrition	Practical: checking prices and deals for yourself.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using × 0.2 for a 20% decrease (or × 0.15 for a 15% increase).
Ask: “Does that give the change, or the new amount?”

Writing a 5% increase as × 1.5.
Ask: “What decimal is 5%? So what is 105%?”

Subtracting the number of percent as pounds (£80 − 35).
Ask: “35% of what? Is that £35?”

Dividing the wrong way round for “as a percentage of”.
Ask: “Which is the part and which is the whole?”

Expecting +10% then −10% to return to the start.
Ask: “10% of which amount is taken off: the old price or the new one?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: fractions, decimals and percentages as one number, and multiplying by a decimal. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says a £240 phone with 15% off costs £36. The aim is a question that brings back the multiplier.

- “What does the £36 tell you?”

- “What percentage of the price do you still pay?”
- “What is 85% as a multiplier?”
- “Does your answer look about right for a small discount?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are calculators allowed?

Yes, from this lesson, as in the iGCSE calculator paper. Learners are asked to estimate first.

Why insist on a single multiplier?

It turns every percentage problem into one calculation and prepares for reverse percentages (dividing by the multiplier) and repeated change (multiplying again), in Lessons 5 and 6.

Is the stretch required?

No. Lesson 6 develops repeated percentage change properly; here it is a first glimpse.

What comes next?

Lesson 4: percentage change, comparing a change with the original amount.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.